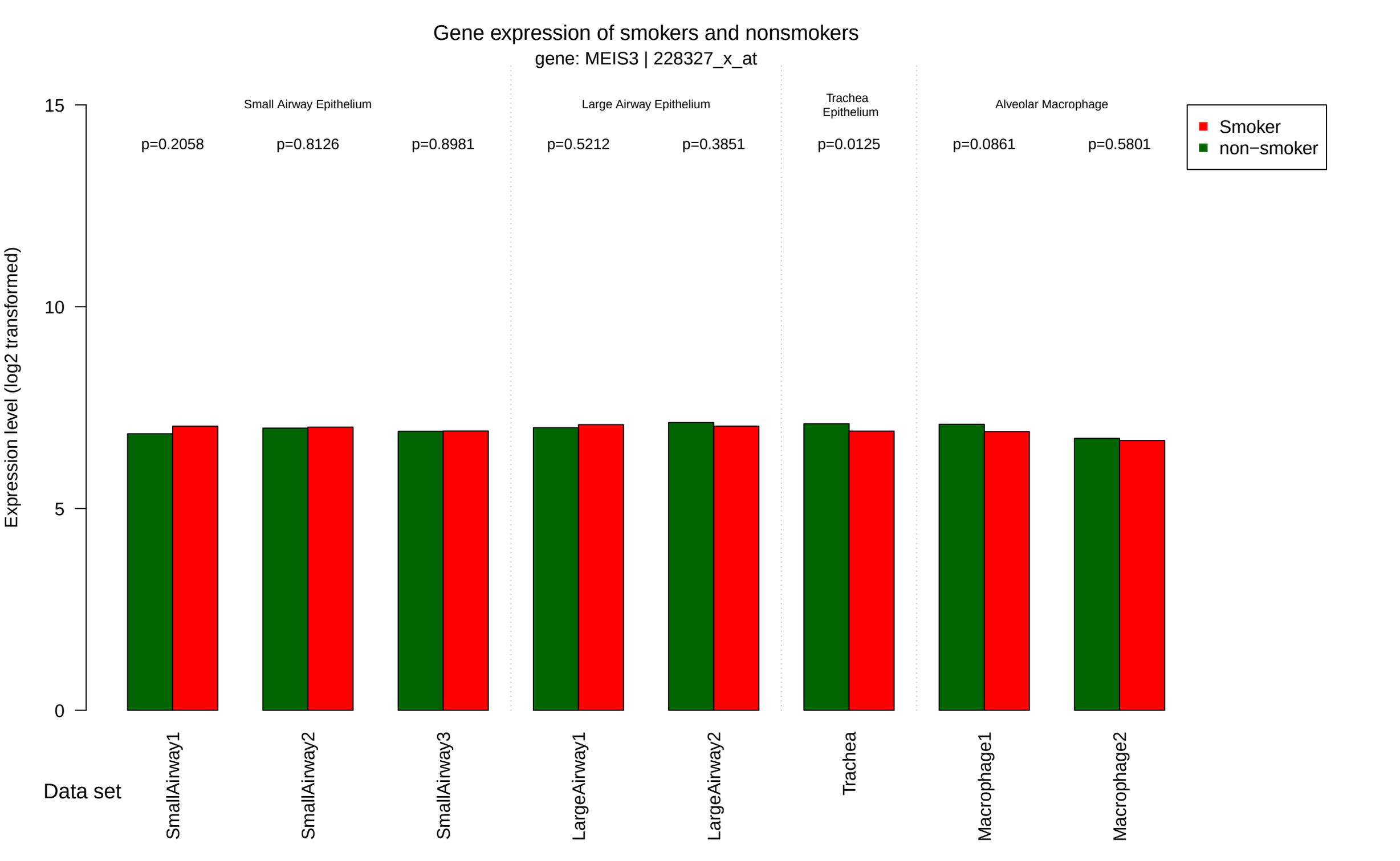
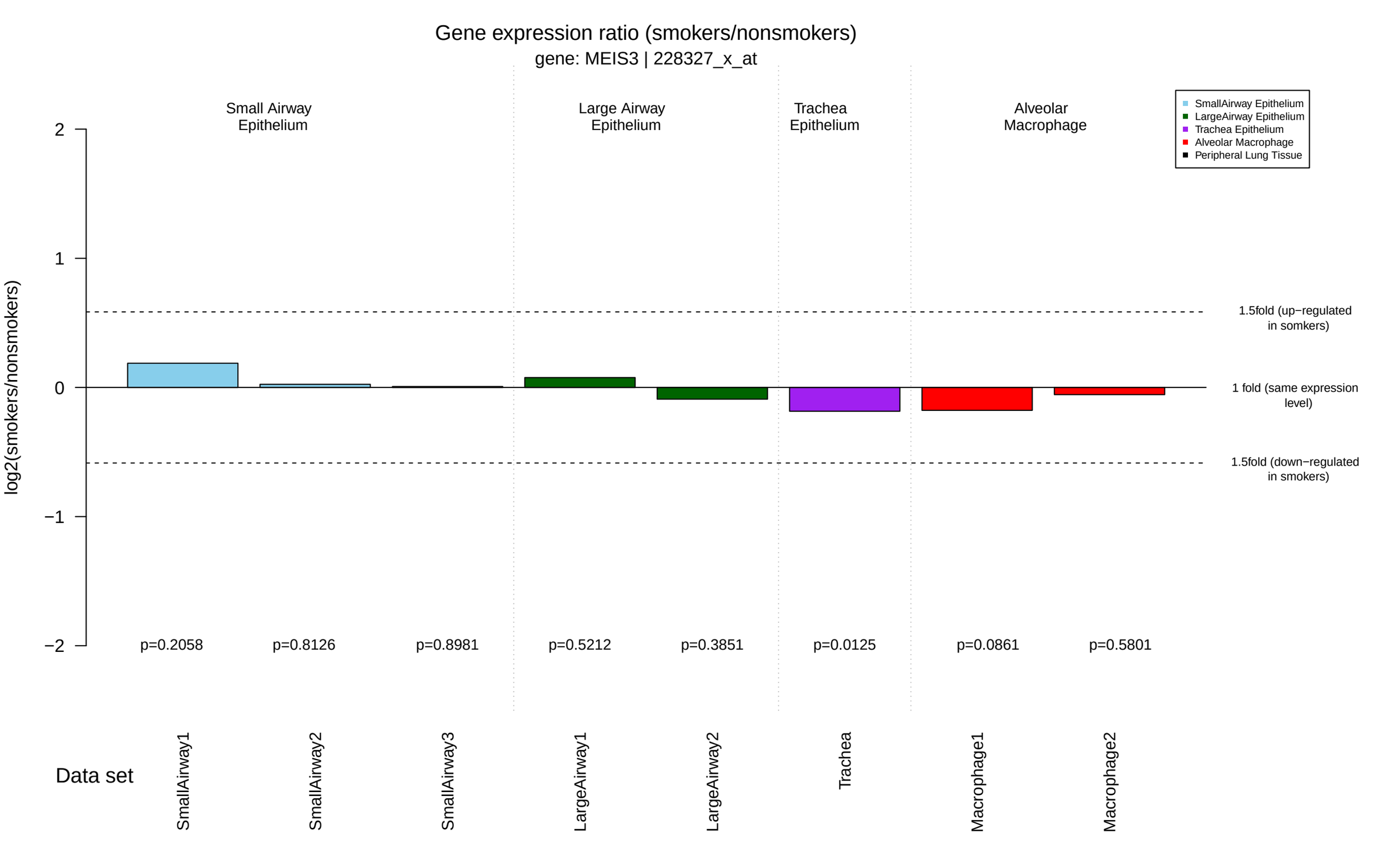
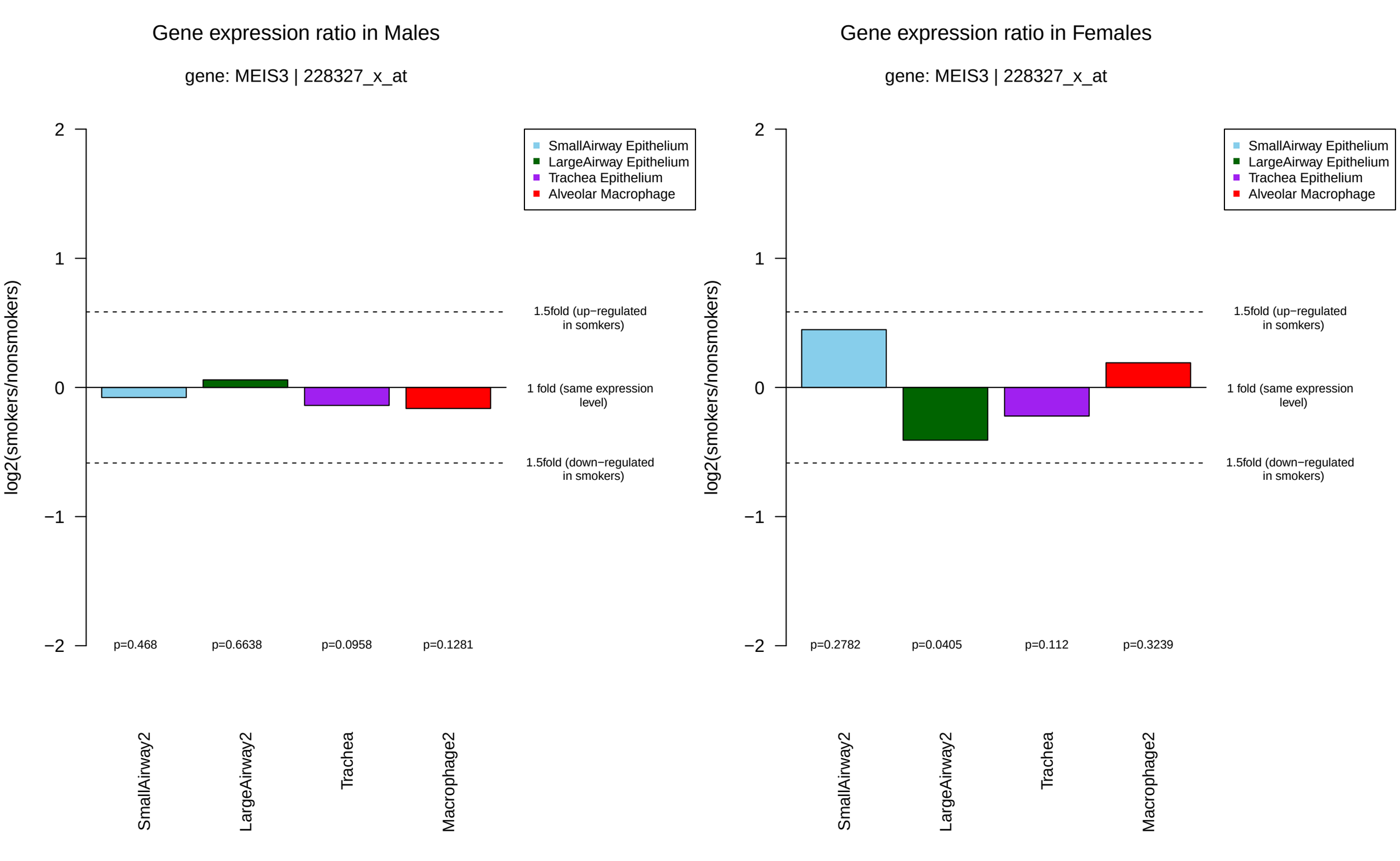
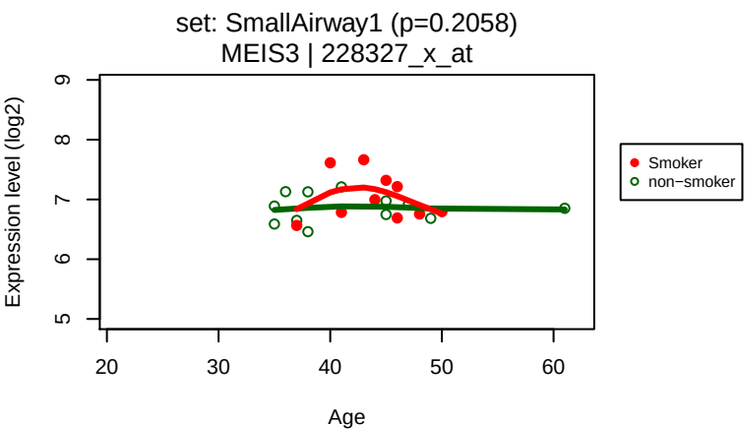
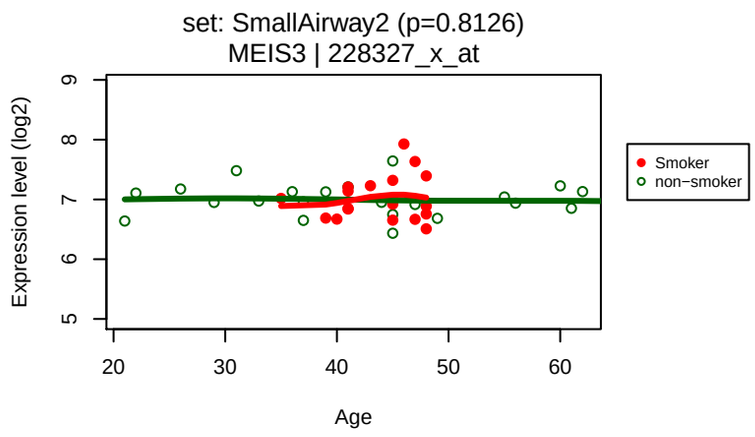
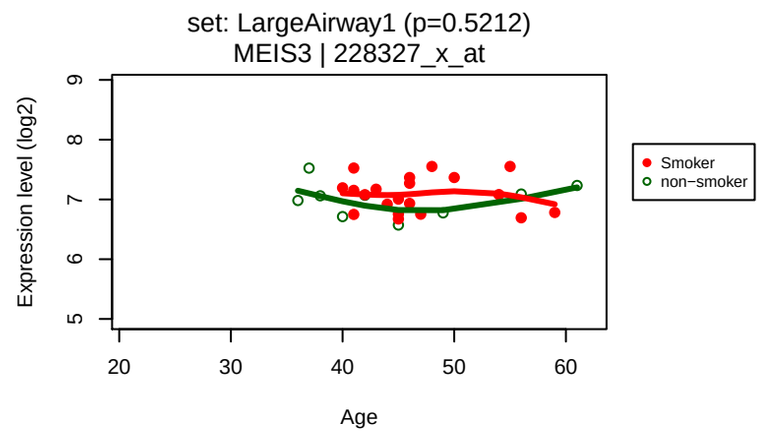

Gene expression of smokers and nonsmokers
gene: MEIS3 | 228327_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2058
p=0.8126
p=0.8981
p=0.5212
p=0.3851
p=0.0125
p=0.0861
p=0.5801
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

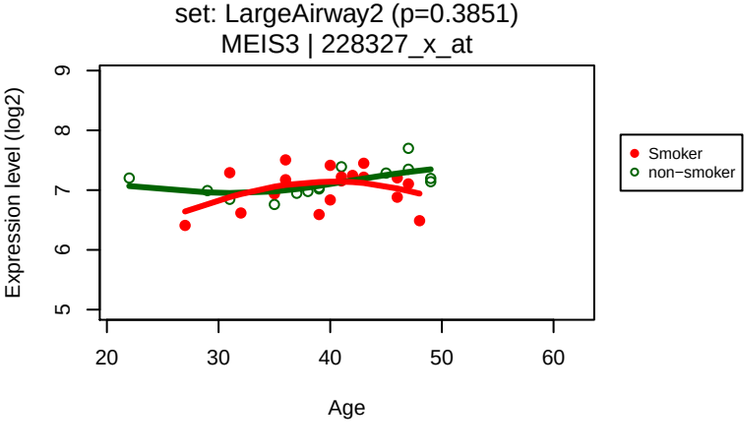

Gene expression ratio (smokers/nonsmokers)
gene: MEIS3 | 228327_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2058
p=0.8126
p=0.8981
p=0.5212
p=0.3851
p=0.0125
p=0.0861
p=0.5801
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

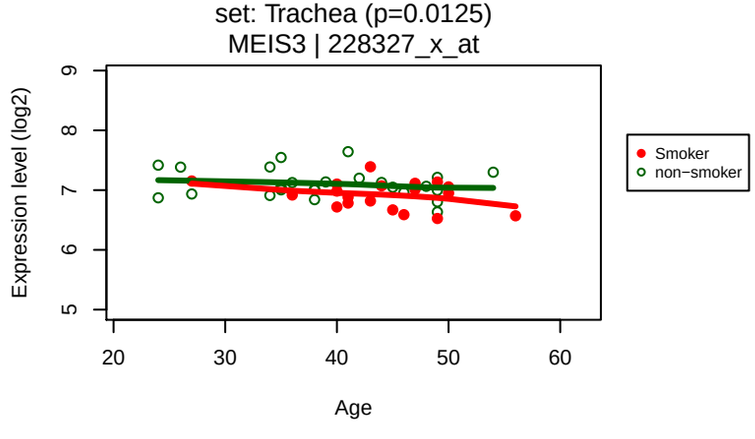

Gene expression ratio in Males
gene: MEIS3 | 228327_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.468
p=0.6638
p=0.0958
p=0.1281
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MEIS3 | 228327_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2782
p=0.0405
p=0.112
p=0.3239
SmallAirway2
LargeAirway2
Trachea
Macrophage2

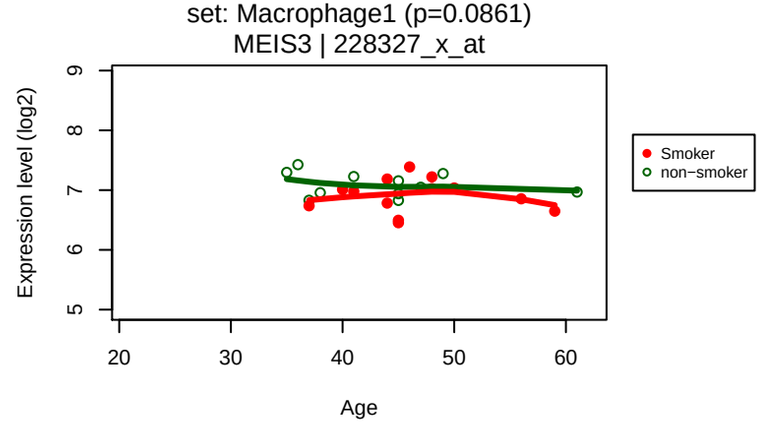

set: SmallAirway1 (p=0.2058)
MEIS3 | 228327_x_at
Expression level (log2)
Age
Smoker
non−smoker

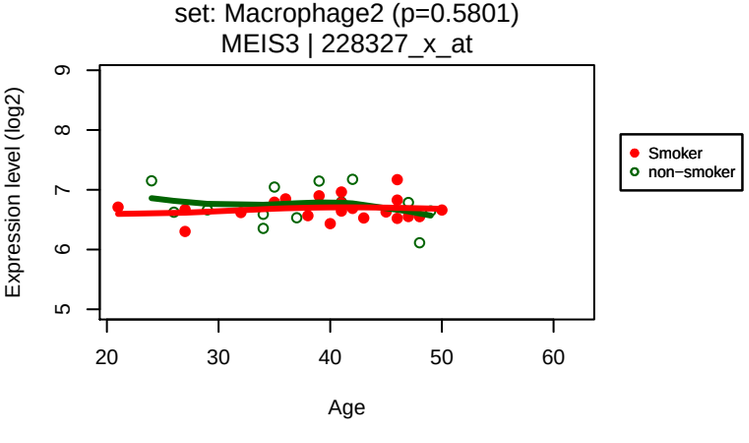

set: SmallAirway2 (p=0.8126)
MEIS3 | 228327_x_at
Expression level (log2)
Age
Smoker
non−smoker

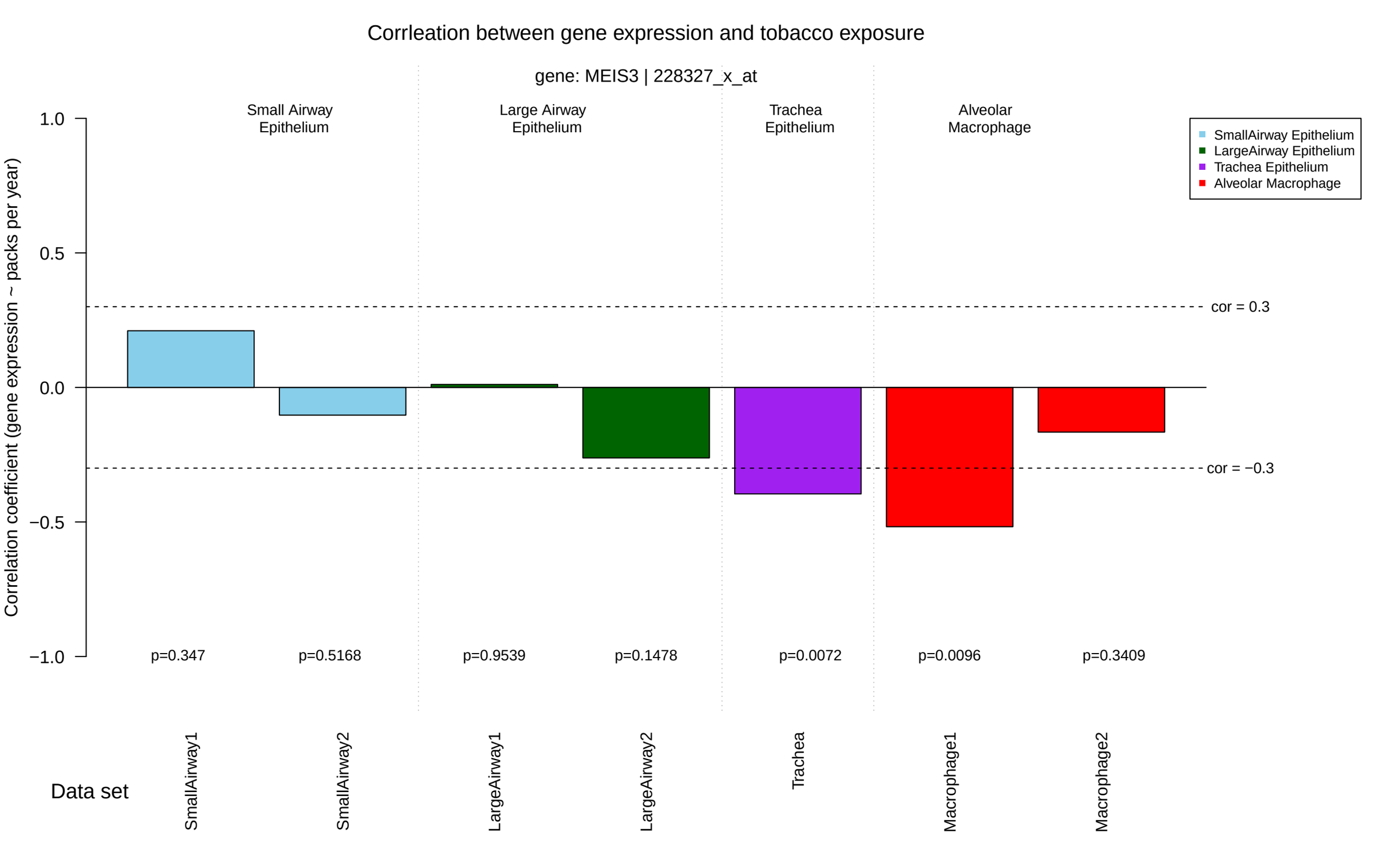

set: LargeAirway1 (p=0.5212)
MEIS3 | 228327_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3851)
MEIS3 | 228327_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0125)
MEIS3 | 228327_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0861)
MEIS3 | 228327_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5801)
MEIS3 | 228327_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MEIS3 | 228327_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.347
p=0.5168
p=0.9539
p=0.1478
p=0.0072
p=0.0096
p=0.3409
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2